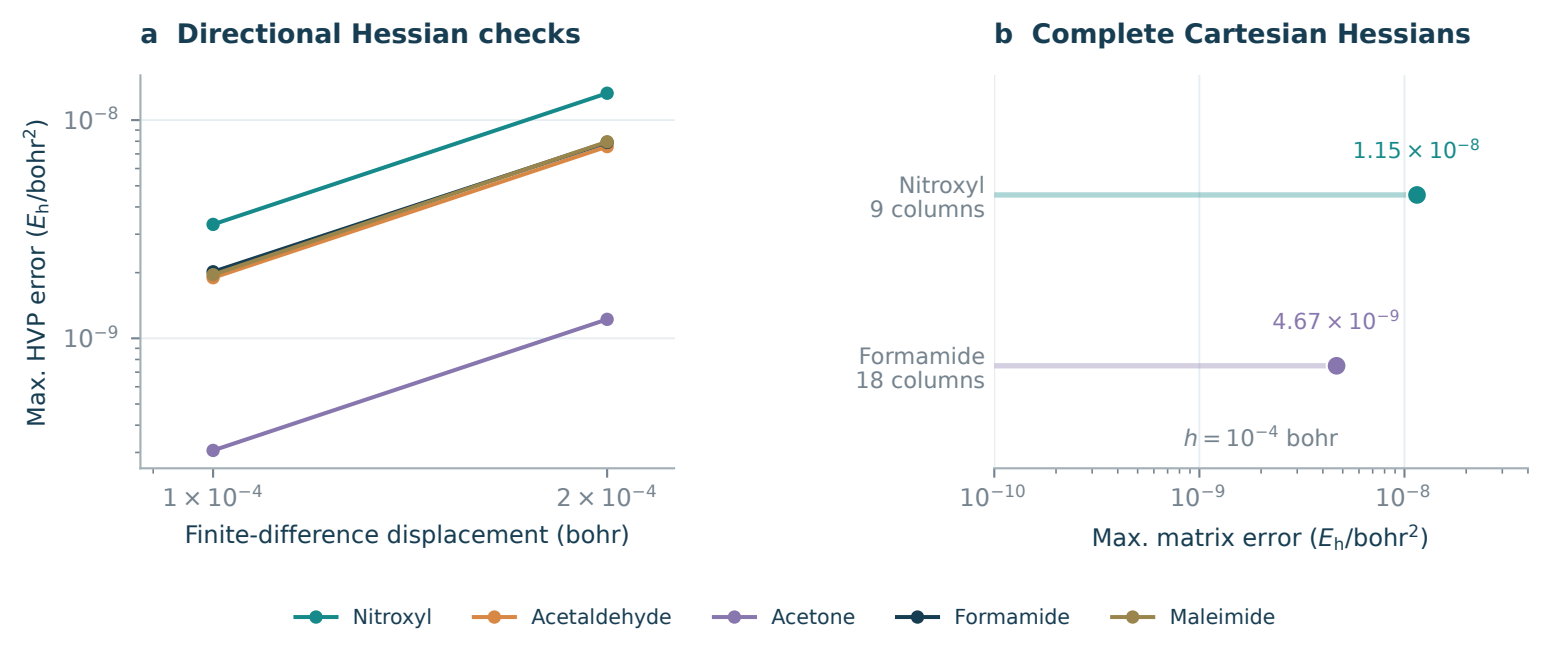
a Directional Hessian checks
Max. HVP error (E_h/bohr^2)
10^{-8}
10^{-9}
1×10^{-4}
2×10^{-4}
Finite-difference displacement (bohr)
b Complete Cartesian Hessians
Nitroxyl
9 columns
1.15×10^{-8}
Formamide
18 columns
4.67×10^{-9}
$h = 10^{-4}$ bohr
10^{-10}
10^{-9}
10^{-8}
Max. matrix error (E_h/bohr^2)
Nitroxyl
Acetaldehyde
Acetone
Formamide
Maleimide